

Quantifying Model, Label, and Methodological Contributions to Variance in Real-Time Sepsis Onset Prediction

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I. Introduction: Discrimination and Clinical Utility

- Sepsis is associated with about **11 million deaths a year**, nearly one fifth of deaths worldwide [1]. Each additional hour before antibiotics is associated with higher mortality [2].
- Wang et al. [3] reviewed **91** real-time sepsis prediction models:

Metric	Internal	External
Median AUROC	0.811	0.783
Median utility score	+0.381	−0.164

- Negative utility indicates performance below the no-alert strategy. AUROC measures ranking, not alert timing or clinical value, and therefore did not reflect the decline in utility.
- Deployment: Epic Sepsis Model v2, 227,091 encounters, AUROC 0.82–0.92, PPV 0.13–0.26 [4].

Model comparisons alone do not establish robustness or clinical utility.

II. Background: Limitations of the Conventional Approach

One label $\rightarrow$ one split $\rightarrow$ AUROC $\rightarrow$ model comparison

1. **Label definition:** applying Sepsis-3 differently produced 867–2,178 cases in the same database [5].
2. **Data split:** sepsis prediction showed the largest performance loss under temporal shift among the clinical tasks evaluated [6].
3. **Metric choice:** high AUROC can coexist with utility below the no-alert strategy [3].
4. **Clinical-action dependence:** labels may capture recognition and treatment practice, not physiological deterioration alone [7].

This thesis

Evaluates model class, label definition, data split, prediction timing and evaluation metric as separate **sources of variation**, with **bootstrap confidence intervals** so that their magnitudes can be compared.

III. Research Question and Pre-Registered Hypotheses

Research Question

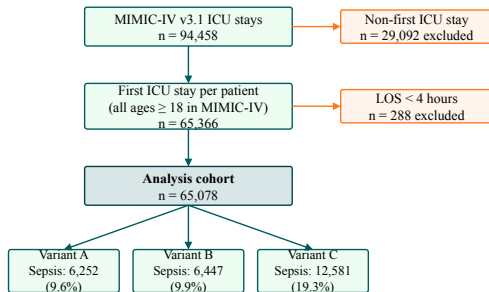
How much of the variation in reported performance comes from the model, and how much from how sepsis is defined, how data are split, when prediction is made relative to treatment, and which metric is reported?

Setting: real-time sepsis detection in MIMIC-IV [8].

- **H1 Label dominance:** label variation $\geq$ model variation in AUROC.
- **H2 Coefficient instability:** ≥ 1 coefficient changes sign or by ≥ 2 -fold across labels.
- **H3 Treatment leakage:** pre-treatment restriction reduces AUROC by ≥ 0.03 .
- **H4 Split optimism:** temporal AUROC is ≥ 0.02 below random-split AUROC.
- **H5 Metric divergence:** utility declines proportionally more than AUROC from internal to external validation.
- **H6 Model-class null:** GBT improves AUROC over the primary model by no more than 0.02.

Multiplicity: Holm correction across H1–H6. **Protocol note:** H2 and H3 yield no p-value; the reported H5 analysis uses a post-hoc substitute estimand. **Estimand:** current-hour onset detection, not advance prediction.

IV. Materials: Data and Cohort



- **Development data:** MIMIC-IV v3.1 [8], single centre (BIDMC, Boston).
- **Cohort:** adults, first ICU stay, length of stay ≥ 4 h; **n** = 65,078.
- **External validation:** eICU-CRD v2.0 [9]; **frozen models, no retraining.**
over 200 US ICUs, 181,589 unit stays
- **Person-hour data:** first 72 h; discharge and death modelled as competing events.
- **22 covariates; 29 model terms per outcome.**
6 vitals, 5 labs, 2 vasopressor flags, 3 six-hour slopes, 6 test-order indicators

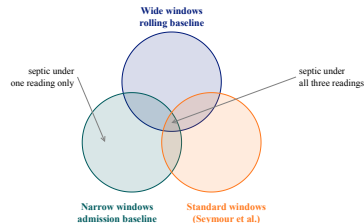
IV. Materials: Three Pre-Registered Sepsis-3 Label Definitions

ID	Name	ABX→cx	cx→ABX	SOFA−	SOFA+	Baseline	Septic stays
A	Narrow	24 h	24 h	24 h	12 h	Admission	6,252 (9.6%)
B	Seymour-Std	72 h	24 h	48 h	24 h	Admission	6,447 (9.9%)
C	Liberal	72 h	72 h	48 h	24 h	Rolling 24 h	12,581 (19.3%)

ABX: antibiotic; cx: culture; SOFA−/SOFA+: permitted hours before/after suspected infection for a ≥ 2 -point SOFA increase.

- All definitions require **suspected infection** (antibiotic paired with a culture) **plus** a ≥ 2 -point SOFA increase [5], [10].
- **B is primary**; A and C are pre-specified alternatives.
- Onset is assigned at suspected infection, t_{susp} . SOFA determines case membership, not onset time.

A and B have similar prevalence but low case agreement: $\kappa = 0.267$.



V. Methods: Models and Validation Architecture

Primary model: dynamic discrete-time competing-risks hazard

Multinomial pooled logistic regression over person-hours (sepsis / discharge / death), with a natural-spline baseline in elapsed hours [11], [12]. A small ridge penalty addresses complete separation.

Comparators: GBT (XGBoost), Cox regression, NEWS2, qSOFA and SIRS [13]. GBT uses fixed hyperparameters and early stopping on a 15% stay-level training holdout.

Split	How	Why
Internal (primary)	Temporal split: train 2008–2016, test 2017–2019 [6]	Primary internal evaluation
Internal (secondary)	Random 75/25 patient-level split (seed 42)	Split optimism (H4)
Treatment-anchored	Hours before first culture or antibiotic [7]	Treatment leakage (H3)
External	eICU-CRD v2.0, no retraining [9]	Generalisability (H5)

All splits are patient-level, with no train–test overlap.

V. Methods: Metrics and Statistical Inference

Primary metric: adapted PhysioNet-style utility score [14]

$$\tilde{U} = \frac{U - U_{\text{inact}}}{U_{\text{opt}} - U_{\text{inact}}} \quad \begin{array}{l} 1: \text{perfect forecaster,} \\ 0: \text{no-alert strategy,} \\ < 0: \text{worse than no alerts} \end{array}$$

- Rewards alerts from 6 h before onset to onset; penalises late and false alerts.
- Reported at the 0.5 threshold **and** at the best point on a 60-point alert-rate grid.
- Alert burden: false alerts per true alert; published benchmark = 1.4 [15].
- AUROC, AUPRC and calibration are reported alongside.

Inference and reporting

- Stay-cluster bootstrap, $B = 2,000$ replicates (1,000 for eICU-CRD).
- Confirmatory family: Holm correction across H1–H6 (FWER 0.05). Exploratory analyses: Benjamini–Hochberg.
- Post-hoc amendments labelled explicitly, including the inference plan itself (Amendment 1); TRIPOD+AI and PROBAST+AI reporting [16], [17].

VI. Results: Internal Temporal Performance (Variant B)

Model	AUROC	95% CI	Utility (best)	Alert burden at best
Primary	0.7590	[0.7434, 0.7749]	0.025	47.3
GBT	0.7424	[0.7265, 0.7583]	0.031	51.8
Cox	0.6444	[0.6271, 0.6614]	0.000	∞
NEWS2	0.6415	[0.6248, 0.6589]	0.000	∞
SIRS	0.6114	[0.5950, 0.6281]	-0.035	135.8
qSOFA	0.6091	[0.5934, 0.6254]	-0.252	248.8

Temporal test set, 2017–2019: 548,827 person-hours and 1,040 onsets. Utility (best): maximised over the threshold grid; ∞ : no alert was optimal. Benchmark burden: 1.4 [15].

- The paired comparison favoured the primary model over GBT by 0.0165 AUROC (H6). Both had higher AUROC than Cox, NEWS2, SIRS and qSOFA.
- Primary-model calibration slope: 1.055 (ideal = 1).
- AUROC in the mid-to-high 0.7s did not translate into a plausible alert burden: the best utility of the primary model and GBT required 47.3 and 51.8 false alerts per true alert.

VI. Results: AUROC Variation Across Analysis Choices

Factor varied (others fixed)	Δ AUROC	95% CI	H
Model class (label B, temporal)	0.1499	[0.134, 0.171]	n/a
Internal – external (primary, label B)	0.0630	[−0.0215, 0.1563]	H5
Label definition (primary, temporal)	0.0123	[0.003, 0.032]	H1
Random – temporal split (primary, label B)	−0.0165	[−0.0372, 0.0040]	H4
GBT – primary (label B, temporal)	−0.0165	[−0.0308, −0.0012]	H6

These are ranges or contrasts, not additive variance components. The model-class range covers six specifications and the label range three definitions, so the two are not directly symmetric. The GBT–Cox difference alone (0.7424 vs 0.6444) is about eight times the full label range.

- **Model class** produced the largest AUROC range.
- The external contrast was second largest but imprecisely estimated: its interval includes zero.
- **Label and split effects were small** in this analysis.

VI. Results: Confirmatory Hypotheses (Holm-Corrected)

H	Contrast	Est.	95% CI	p_{Holm}	Verdict
H1	Label spread — model spread	−0.1375	[−0.1623, −0.1083]	1.000	NOT confirmed
H2	Unstable covariates (count)	6	n/a	n/a	Descriptive
H3	Pre-treatment onsets (count)	0	n/a	n/a	Definitional
H4	Random — temporal AUROC	−0.0165	[−0.0372, 0.0040]	1.000	NOT confirmed
H5 [†]	Internal — external AUROC	0.0630	[−0.0215, 0.1563]	0.551	Not evaluable as pre-registered; substituted AUROC contrast inconclusive
H6	GBT — primary AUROC	−0.0165	[−0.0308, −0.0012]	1.000 [‡]	H6 not rejected

[†] Post-hoc substitute: internal utility was too close to zero for a proportional decline to be defined, so an AUROC contrast takes its place. [‡] Tests the pre-registered interval null $|\theta| \leq 0.02$; non-inferiority is read from the confidence interval, not from this p-value.

- **H1 not confirmed:** model spread 0.1499 vs label spread 0.0123; opposite direction, interval excludes zero.
- **H4 not confirmed:** random-split AUROC was 0.0165 *lower* than temporal (0.7424 vs 0.7590); the interval includes zero.
- **H6:** GBT — primary = −0.0165; the upper limit (−0.0012) lies below the +0.02 margin, supporting one-sided **non-inferiority**, not two-sided equivalence.

VI. Results: Why H3 Was Not Evaluable

Status: (NOT EVALUABLE AS PRE-REGISTERED). The pre-registered onset rule produced no pre-treatment onsets, so the contrast could not be formed.

Structural reason

In Variants A, B, C and E, onset is assigned to the *later* event of the antibiotic–culture pair, so onset cannot precede the treatment anchor. The window is empty **by definition**, not because no events were observed.

Var	Label	Pre-treatment person-h	Onsets	AUROC (primary / GBT)
B	Seymour-Std	179,003	0	undefined
F	Sepsis-3 Early-Onset	178,303	71	0.8081 / 0.8018
D	Deterioration	147,299	1,362	0.8498 / 0.9360

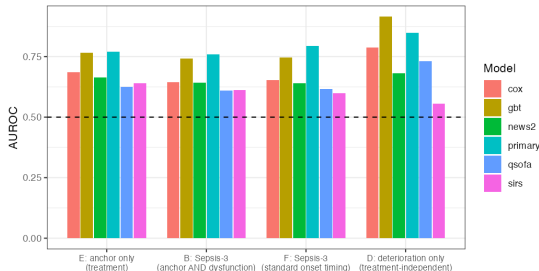
- **Variant F** (post-hoc) keeps B's cases but uses the standard earlier onset, $\min(t_{\text{susp}}, t_{\text{SOFA}})$. It moves 1,738 of 6,447 onsets earlier (median shift -14.0 h), 71 of them into the test-set pre-treatment window.
- The concern about treatment-dependent case membership [7] remains relevant; its effect on onset timing depends on the onset rule. These AUROCs assess detection at the assigned onset, not advance prediction.

VI. Results: Post-Hoc Decomposition of the Sepsis-3 Label

Labels D, E and F are post-hoc protocol amendments: (EXPLORATORY, POST-HOC)

Splitting the Sepsis-3 conjunction into its two limbs

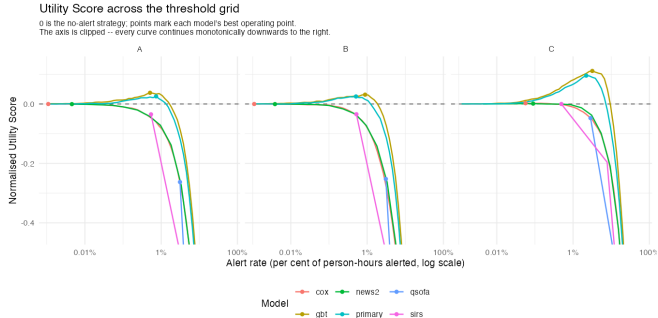
Temporal test set, unrestricted window; dashed line = chance



- **E (Anchor-Only)** removes SOFA but keeps B's onset hour in 100.0% of B's cases: SOFA affects membership, not onset time.
- Anchor-attributable fraction ϕ_{anchor} : primary 1.041 [0.990, 1.097]; GBT 1.099 [1.037, 1.164].
- **D (Deterioration)** removes treatment timestamps; agreement with B is low ($\kappa = 0.057$). Five of its six SOFA components are model inputs, so its AUROC is partly self-fulfilling.
- **F (standard onset timing)**: primary AUROC 0.7590 $\rightarrow$ 0.7951; the GBT change was small and not statistically detectable ($\Delta = 0.0033$, $q = 0.559$).

For the primary model, onset timing had a larger effect than removing SOFA. No corresponding timing effect was detected for GBT.

VI. Results: Clinical Utility Across Alert Thresholds



- At the 0.5 threshold the primary model issued no alerts, so its utility was 0.000.
- Across all models and thresholds, the highest Variant B utility was 0.031 (**GBT**), at 51.8 **false alerts per true alert** (benchmark: 1.4 [15]); under any pre-registered variant, 0.111 (GBT, C).
- Thresholds were selected on the test set, so these maxima are optimistic.

VI. Results: External Validation Limited by Label Observability

Var	Internal (primary)	eICU primary	eICU GBT	eICU NEWS2	eICU events
A	0.7713	0.6913	0.7106	0.5866	27
B	0.7590	0.6960	0.6815	0.5259	29
C	0.7695	0.7370	0.6877	0.5074	73

eICU microbiology-covered subset; frozen models, no retraining.

- Culture data covered **1.55% of eICU cohort stays**, leaving 29 Variant B events, below the 100-event interpretability floor.
- H5 substitute, internal – external AUROC = 0.0630 [−0.0215, 0.1563]: the point estimate is lower externally, but the interval includes zero, so the contrast is inconclusive. The source of the difference was not identified.
- Antibiotic-only (6,140 events) and Variant D (52,535 events) analyses had more events but measure different targets; both showed lower external discrimination.

VI. Results: Coefficient Instability and Subgroup Analysis

H2: coefficient instability (descriptive)

- 6 of 22 covariates met the instability rule: 2 by sign, 5 by ≥ 2 -fold magnitude (categories overlap).
- **After accounting for estimation uncertainty, 3 remained unstable, all test-order indicators [18].** The raw count is an upper bound on the instability present.
- **Ablation** (post-hoc): removing the 8 action-derived covariates reduced instability from 6/22 to 1/14 and changed AUROC by -0.0076 [$-0.0131, -0.0022$] (primary, B): small, but not absent.

Label sensitivity by recorded subgroup (exploratory)

- Subgroup AUROC: 6 contrasts met the 100-event requirement; **none remained significant** after BH correction.
- Only two race categories met it, one of them *Unknown*, so no adequately powered comparison between two recorded categories was available.
- Label sensitivity: the only racial strata above the White reference are *Unknown* ($+3.01$ pp, $q = 0.006$) and *Unable to obtain* ($+3.49$ pp, $q = 0.009$), both **missingness categories**.
- This may reflect record completeness, a mechanism not tested [19]; no stratum recording an actual answer differs from the reference.

VII. Discussion: Main Findings and Limitations

Main findings

1. **Treatment-dependent membership and onset timing:** the pre-registered onset rule made H3 non-evaluable; standard timing moved 1,738 onsets earlier.
2. **Definition sensitivity:** prevalence ranged from 9.6% to 19.3% and case agreement was low ($\kappa = 0.267$), but the AUROC range was only 0.0123.
3. **Limited clinical utility:** the best Variant B operating point required 51.8 false alerts per true alert (benchmark 1.4).

Improving discrimination alone cannot resolve limitations introduced by outcome membership, onset timing and alert burden.

Limitations

- Current-hour onset detection, not advance prediction.
- External Variant B analysis: 1.55% culture coverage, 29 events.
- Single-centre development; first ICU stays of at least 4 h.
- The pre-registered onset rule differs from the standard rule; Variant F restores standard timing post hoc.
- Variant D is an unvalidated label whose components overlap the model inputs.
- Post-hoc analyses are labelled; the literature review is narrative.

VIII. Conclusion: Contributions and Future Work

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- C1 Model class produced the largest AUROC range; label definition changed AUROC by only 0.0123 despite low case agreement ($\kappa = 0.267$).
 - C2 The pre-registered onset rule leaves the pre-treatment window structurally empty; standard timing places 71 test-set onsets there.
 - C3 Maximum Variant B utility was 0.031, requiring 51.8 false alerts per true alert.
 - C4 3 covariates remained unstable after accounting for estimation uncertainty, all test-order indicators.
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Also: C5 significant label-sensitivity contrasts involve missingness categories; C6 pre-registered hypotheses, disclosed amendments, a fixed seed and values written mechanically from result tables; C7 definitional versus empirical label variation ($\kappa_{D,B}$, ϕ_{anchor}).

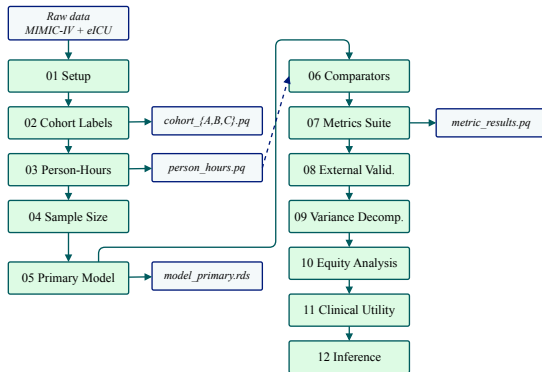
Future work: labels less dependent on treatment timestamps and adjudicated independently of the model inputs; lagged predictors; SEP-1 and CDC Adult Sepsis Event definitions; prospective multi-centre validation; utility-optimised training.

The main effect of label choice was on cohort membership and onset timing rather than on AUROC. Model choice affected AUROC more, and utility stayed close to the no-alert strategy.

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Appendix: Reproducible Analysis Pipeline



- One orchestration script runs all 12 stages.
- One configuration file defines every path, identifier, window, threshold and seed.
- Every reported value is written mechanically from the pipeline's result tables; none is transcribed by hand.
- Frozen run: frozen-2026-08-31.
- Code:

github.com/ahadarif007/sepsis-variance-decomposition

Thank you

Questions and discussion

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Code: `github.com/ahadarif007/sepsis-variance-decomposition`